motion changes. Similarly, future drone missions planned for Titan, Saturn's largest moon, will leverage its dense atmosphere and low gravity to enable highly energy-efficient flight. NASA's upcoming Dragonfly mission exemplifies this approach, using a dual-rotor octocopter to explore Titan's organic-rich landscapes and study prebiotic chemical processes.

Beyond planetary surfaces, drones designed for asteroids and orbital infrastructure will rely on micro-propulsion systems, cold-gas thrusters, ion engines, and robotic manipulators to maneuver in microgravity environments. They may be deployed to survey mineral-rich asteroids, support in-space construction, or assist in assembling solar power or fuel processing stations. These missions require drones to be radiation-hardened, autonomously intelligent, and capable of operating with intermittent communication links, since human command inputs are delayed over interplanetary distances.

The development of drones for space exploration thus represents a multidisciplinary engineering frontier, integrating aerospace design, robotics, artificial intelligence, astrobiology, and planetary science. As these technologies mature, drones will become essential partners in expanding human presence beyond Earth, enabling exploration, science, and early-stage infrastructure for future settlements on Mars, the Moon, and deep-space environments.

9.7.3 Drones in space mining and infrastructure development

As space agencies and private companies move toward establishing a sustained human and industrial presence beyond Earth, drones are expected to play a central role in space mining, resource assessment, and infrastructure development on the Moon, Mars, and near-Earth asteroids. In these environments, drones offer the ability to survey wide areas rapidly, identify resource deposits such as water ice, metallic ores, and volatile